Stat → Day 1

Types of Statistics
→ Descriptive Statistics
→ Mean, Median, Mode
→ Variance and Standard Deviation
→ Histogram, PDF, PMF

→ Internal Statistics
→ Using the data we have collected to form conclusions.
→ Z test, t test, Chi Square test
(Hypothesis testing)

Population & Sample Data

Population
→ The whole group of data

Sample Data
→ The subset of a population.

Types of Data.

1) Discrete Data ⇒ Discrete data are countable values with no values in between.

2) Continuous Data ⇒ Continuous data are measurable values that can take any values in a range, including fractions and decimals.

⇒ Key points of Discrete Data
- → Always Whole Number
- → Cannot be divided into small parts manually

Eg:- No. of students in a class ⇒ 25 not 25.5
No. of cars parked ⇒ 5 not 5.5
No. of siblings you have ⇒ 2 not 2.5

⇒ Key points of Continuous Data
- → Comes from Measurement
- → Can have infinite values within a range
- → Values can be divided further

Eg:- Height of person → 5.2 m
Temperature ⇒ 32.5°C
Time ⇒ 2.5 hours

Scale of Measurement 1) Nominal Scale Data
→ Ranking not Important
→ For eg:- Gender, Color

2) Ordinal Scale Data
→ Ranking is Important
→ Difference cannot be measured
→ For eg:- worst, worse, Bad, Good, Better, Best
→ For eg:- 1 star, 2 star

3) Interval Scale Data
→ Ranking is Important
→ Difference can be measured
→ Ratio cannot be measured (weight, Age)
→ For ex:- Temperature (-30F, -10F, 30F, 60F)
→ starting point should not be 0 (weight, Age)

4) Ratio Scale Data
→ Ranking is important
→ Difference can be measured
→ Ratio can be measured
→ Can contain 0 as starting point
→ For eg:- students marks in a class

Mean => Average   np.Mean (age)

Median => Sort the elements
=> Count the no. of elements
=> if count == even then Median = sum of mid/2
=> If count == odd then Median = mid
np. median (weight)

When to use median?
=> Median can be used if there is outlier present

Mode :- The highest repeated data in the dataset
Mode is not found directly in numpy so we need to use scipy library.

from scipy import stats
stats. mode (age)

(XI) Variance

→ Variance tell us how far the numbers are from the average.

Suppose,

marks: [1, 2, 3] [2, 4, 6]

Mean: 2 4

np.var(a8)

Variance : $\frac{(2-1)^2+(2-2)^2+(2-3)^2}{3-1}$ $\frac{2^2+0^2+(-2)^2}{3}$

: $\frac{1^2+0^2+(-1)^2}{3}$ $\frac{8}{3}$

: $\frac{2}{3}$: 2.67

For sample, N-1 to create unbiased estimator of population variance.

(XII) Standard Deviation

→ Standard Deviation is simply a sq. root of variance.

np.std(age)

Corelation & covariance → Determines the relation between the tables.

df.var() ⇒ Provides variance of row ⇒ df.var(axis=0)
df.var(axis=1) ⇒ Gives variance of column

Random Variable
→ Process of mapping of output of random process to a number

Histogram and Skewness

Histogram
→ Special kind of bar chart that shows how many times a value or range of value appears.

Skewness
→ Skewness tells us if the data is balanced or leaning more to one side.

3 types of Skewness
1) Symmetric → like a hill → left = Right
Positive 2) Right Skewed → Tail is longer on right → Mean > median
Negative 3) left Skewed → tail goes left → Mean < median

(8) Normal Gussian Distribution

→ Normal Gussian distribution is a bell shaped curve that shows most values are in the center around the average.

(9) Covariance and Correlation

→ Determines the relation between the tables.

Four conditions:-

1) when x increases, y increases
2) when x decreases, y decreases
3) when x increases, y decreases
4) when x decreases, y increases

(10) ~~Pearson Correlation Coefficient~~

Covariance → It tells us how two variables changes together. Value can be any n

Correlation → It tells us how perfectly they are related (from -1 to +1).

Covariance Formula

$$Cov(X,Y) = \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})$$

X_i & Y_i are values
$\bar{X}, \bar{Y}$ are means
n is no. of data points

Correlation Formula

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sigma_x \cdot \sigma_y}$$

σ_x, σ_y = Standard deviation of x and y.

Always gives a value between -1 and 1.

Eg Marks of 2 students in 3 tests

Test	X	Y
1	60	70
2	70	80
3	80	90

$$\bar{x} = \frac{60+70+80}{3} = 70$$

$$\bar{y} = \frac{70+80+90}{3} = 80$$

$$\text{Cov}(x,y) = \frac{1}{3}[(60-70)(70-80) + (70-70)(80-80) + (80-70)(90-80)]$$

$$= \frac{1}{3}[(-10)(-10) + 0 + (10)(10)]$$

$$= \frac{1}{3}[100+0+100]$$

$$= \frac{200}{3} \approx 66.67$$

Covariance is positive, so marks goes up together.

Correlation

Fr correlation, first calculate standard deviation

$$\sigma_x = \sqrt{\frac{(-10)^2 + 0^2 + 10^2}{3}} = \sqrt{\frac{200}{3}} \approx 8.16$$

$$\sigma_y = \sqrt{\frac{(-10)^2 + 0^2 + 10^2}{3}} = \sqrt{\frac{200}{3}} \approx 8.16$$

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sigma_x \sigma_y} = \frac{66.67}{8.16 \times 8.16} \approx 1$$

$\therefore$ Since, correlation = 1, X and Y have perfect positive relation.

Correlation	Meaning
+1	Perfect Positive
0	No relation
−1	Perfect Negative
~0.5	Moderate Positive
~−0.5	Moderate Negative

So, this is Pearson Correlation Coefficient. When it is correlation, it usually means Pearson Correlation Coefficient

(4) Spearman Rank Correlation

- It checks how well the relationship between two variables can be described using a monotonic function

It uses ranks instead of actual data values

Formula

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$$

where,
d_i = difference between the ranks of corresponding values

n = no. of data points

Eg $X = [10, 20, 30, 40, 50]$
$Y = [100, 80, 60, 40, 20]$

Step 1: Rank the values of X and Y.

X	Rank(X)	Y	Rank(Y)
10	1	100	5
20	2	80	4
30	3	60	3
40	4	40	2
50	5	20	1

Step 2:- Find difference in ranks (d)

Rank(X)	Rank(Y)	d = Rank(X) - Rank(Y)	d^2
1	5	-4	16
2	4	-2	4
3	3	0	0
4	2	2	4
5	1	4	16

$\therefore \sum d^2 = 16 + 4 + 0 + 4 + 16 = 40$

$$\therefore \rho = 1 - \frac{6\sum d^2}{n(n^2-1)} = 1 - \frac{6 \times 40}{5(25-1)} = 1 - 2 = 1$$

Eg $X = [10, 20, 30, 40, 50]$
$Y = [20, 40, 60, 80, 100]$

X	Rank(X)	Y	Rank(Y)
10	1	20	1
20	2	40	2
30	3	60	3
40	4	80	4
50	5	100	5

Rank(X)	Rank(Y)	d	d^2
1	1	0	0
2	2	0	0
3	3	0	0
4	4	0	0
5	5	0	0

$\therefore \rho = 1 - \frac{6 \times 0}{5(25-1)} = 1 - 0 = 1$

$\therefore$ There is perfect positive correlation between X and Y.

(7) Implementation Through Python

```
import seaborn as sns
df = sns.load_dataset('healthexp')
df.head()

numeric_df = df.select_dtypes(include=['number'])
numeric_df.head()

numeric_df.corr(method='pearson')

numeric_df = df.select_dtypes(exclude=['number'])
numeric_df.head()

numeric_df.cov()

numeric_df.corr(method='spearman')
```

Kendall's Tau

→ Kendall's tau measures the strength and direction of association between two ranked variables, just like Spearman but with different method.

Formula,

$$\tau = \frac{(C-D)}{\frac{n(n-1)}{2}}$$

where,

C = No. of concordant pairs
D = No. of discordant pairs.
n = No. of data pairs
$\frac{n(n-1)}{2}$ = Total no. of pairwise comparisons

Eg:- X = [10, 20, 30, 40]
Y = [30, 60, 20, 90]

Paired Comparison	X change	Y change	C/D
(10,20)-(30,60)	↑	↑	C
(10,30)-(30,20)	↑	↓	D
(10,40)-(30,90)	↑	↑	C
(20,30)-(60,20)	↑	↓	D
(20,40)-(60,90)	↑	↑	C
(30,40)-(20,90)	↑	↑	C

$\therefore$ Concordant (C) : 4 $\quad\quad$ $\therefore$ Total Pairs : 6
Discordant (D) : 2

$$\therefore \quad \tau = \frac{C-D}{\text{Total Pairs}} = \frac{4-2}{6} = \frac{2}{6} = 0.333$$